

Shrub Hub

MAKER GUIDE

Overview

This document contains the necessary information to build the Shrub Hub, a low cost, easy to build digital switch interface.



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Maker Checklist

This list provides an overview of the steps required to build and deliver the Shrub Hub.

Maker To Do List

- ☐ Read through the Maker Guide to become familiar with required components, tools, supplies, safety gear, and overall assembly steps.
- ☐ Talk to the User about customization options
 - Colour of printed enclosure
 - How they would like to receive the “User Guide” (PDF or physical copy)
- ☐ Order hardware components
- ☐ Gather tools, supplies, and safety equipment.
- ☐ Assemble the device
- ☐ Flash firmware to Shrub Hub
- ☐ Test the Shrub Hub
- ☐ Print “User Guide” (if the User would like a physical copy)

Items to Give to User

- ☐ Shrub Hub
- ☐ “User Guide”

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Tool List

Tools / Equipment

Tool ID	Description	Required / Recommended	Notes
T01	Wire Strippers	Recommended	Stripping mono cable wires.
T02	Small Phillips head screwdriver	Recommended	Operating TRRS terminal screws
T03	Silver Sharpie	Recommended	Marking each cable

Customization Guide

The device can be printed in the user's desired colour.

3D Printing Guide

The device was originally printed on a Bambu P1S using BambuStudio.

3D Printing Summary

Metrics	Single Unit
Total Print Time (hour min)	0:25
Total Number of Components	4
Typical Total Mass (g)	10
Typical Number of Print Setups	1

3D Printing Settings

Note that the 3D printing material should be assumed to be PLA unless otherwise noted in the table below.

Print File Name	Qty	Total Print Time (hr:min)	Mass (g)	Infill (%)	Support(Y/N)	Layer Height/ Nozzle Diameter(mm)	Notes
Shrub Hub Base.stl	1	0:06	3	20	N	0.2/0.4	
Shrub Hub Lid.stl	1	0:05	2	20	N	0.2/0.4	
Strain Relief Base.stl	1	0:08	3	20	N	0.2/0.4	
Strain Relief Top.stl	1	0:06	2	20	N	0.2/0.4	

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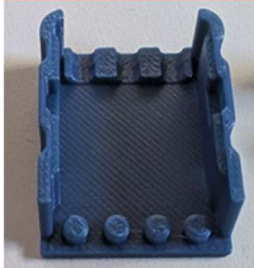
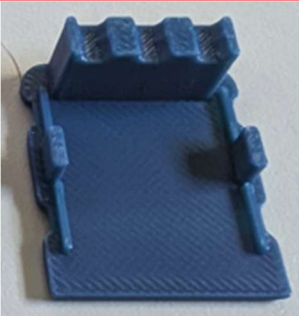


Post-Processing

Inspect the 3D printed parts for any printing defects, sharp edges, or burrs. Sharp edges and burrs can be removed with sanding or deburring tools.

Examples of Quality Prints

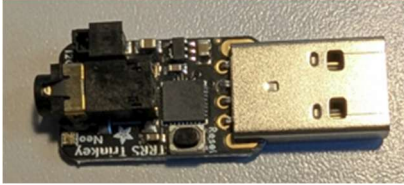
Compare your 3D prints to the images here. If there are significant differences, you may need to reprint the part.

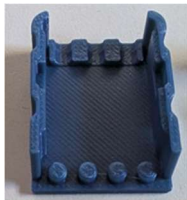
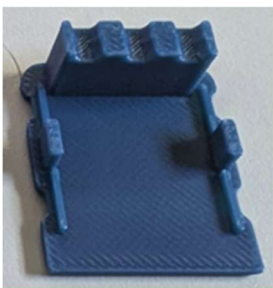
Shrub Hub 3D Prints		
Shrub Hub Base.stl	Shrub Hub Lid.stl	Strain Relief Base.stl
		
Strain Relief Top.stl		
		

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Maker Component List

Part A Components								
A01	TRRS Trinkey	QTY: 1	A02	Shrub Hub Base.stl	QTY: 1	A03	Shrub Hub Lid.stl	QTY: 1
								
A04	Light Pipe	QTY: 1						
								

Part B Components								
B01	TRRS Terminal	QTY: 1	B02	Female Mono Cable	QTY: 3	B03	Strain Relief Base	QTY: 1
								
B04	Strain Relief Top	QTY: 1	B05	Cable Tie	QTY: 2			
								

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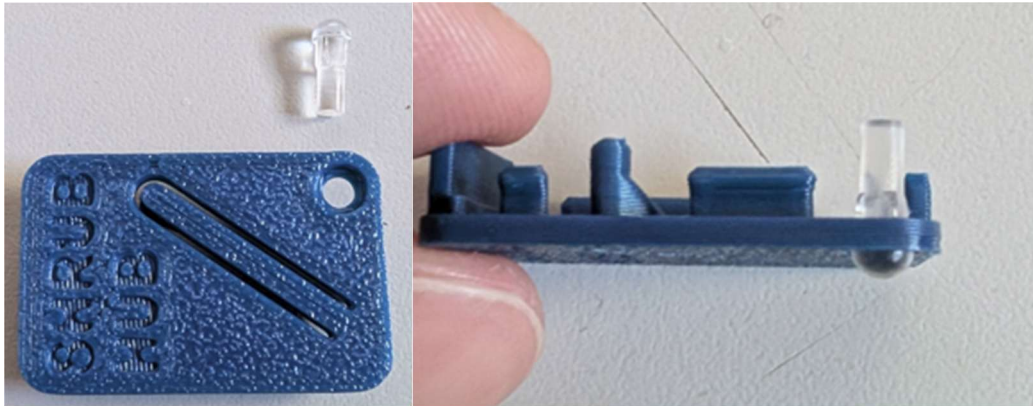
Assembly Guide

Part A: Assembling the Shrub Hub

Part A: Shrub Hub Assembly Steps

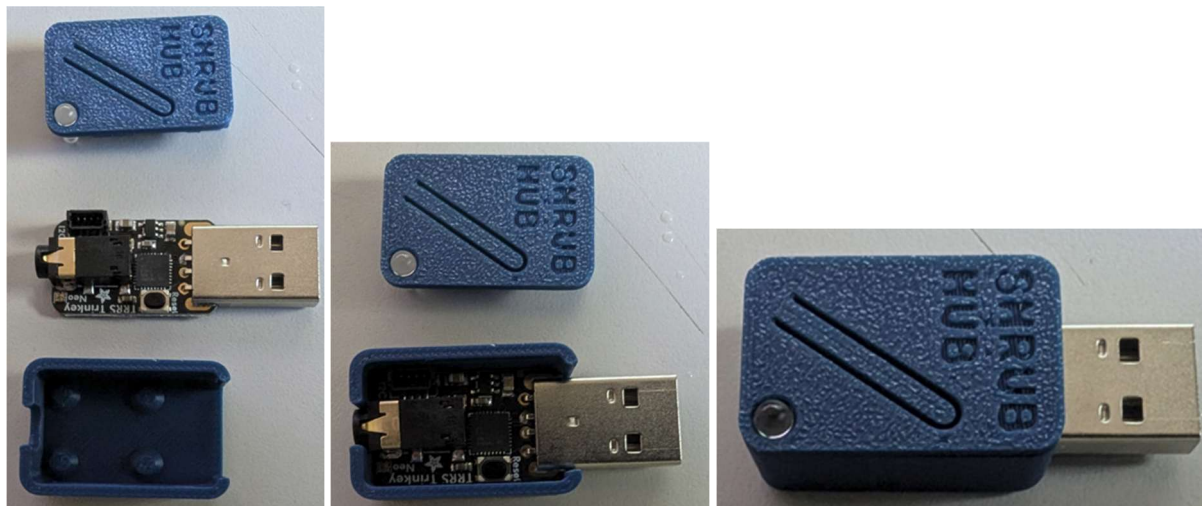
Step A-01: Attach the enclosure to the Trinkey

Take the light pipe (A04) and press it into the hole on the enclosure top(A03).



Step A-02: Attach the enclosure to the Trinkey

Place the TRRS Trinkey (A01) into the base of the enclosure(A02), then snap the top of the enclosure(A03) on top of it to lock the Trinkey in place.



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Part B: Assembling the Cable Splitter

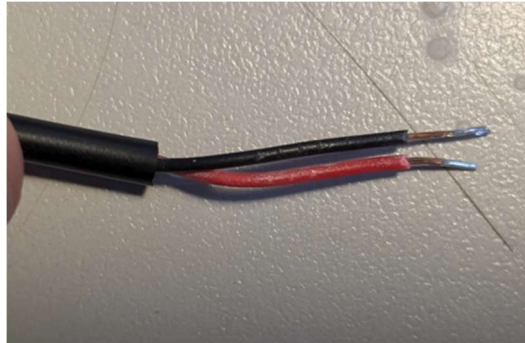
Part B: Required Tools and Supplies

- Small Phillips screwdriver
- Wire Strippers
- Silver Sharpie

Part B: Cable Splitter Assembly Steps

Step B-01: Strip the mono cables

Using wire strippers (T01), strip the ends on each mono cable(B02) so there is 1cm of wire exposed. The wires do come pre stripped, but having slightly more wire available makes the next step easier.

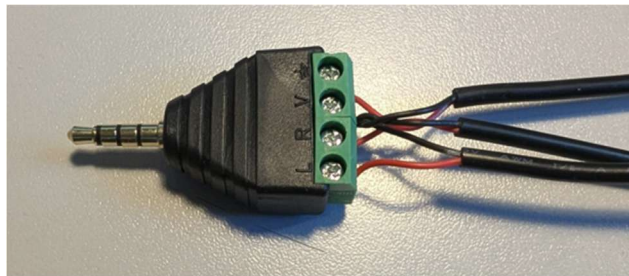
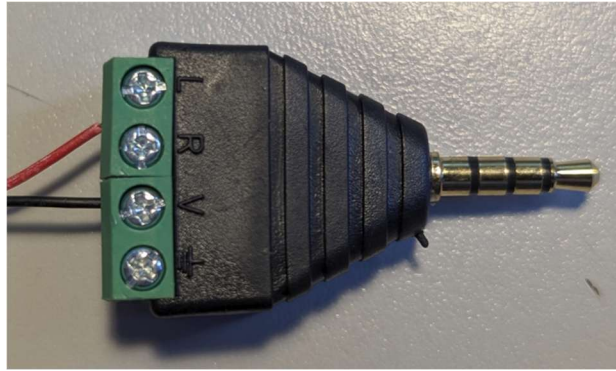


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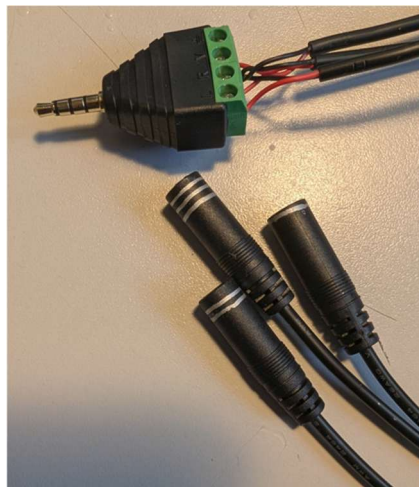
Step B-02: Connect the mono cables to the terminal

Using a small Phillips screwdriver (T02), open the terminals on the TRRS terminal jack(B01). Put the black wires from all three of the stripped mono cables into the terminal marked V and then tighten the terminal to lock all the wires in place. Take the red wire from each of the mono cables attach one to each of the remaining terminals, one to L, one to R, and one to the ground.



Step B-03: Mark the cable connectors

Using a silver sharpie (T03) or other marking device, mark the cable plug for switch 1, switch 2, and switch 3. Switch 1 is the one with the red wire connecting to the L terminal, switch 2 connects to the R terminal, and switch 3 connects to the ground terminal.

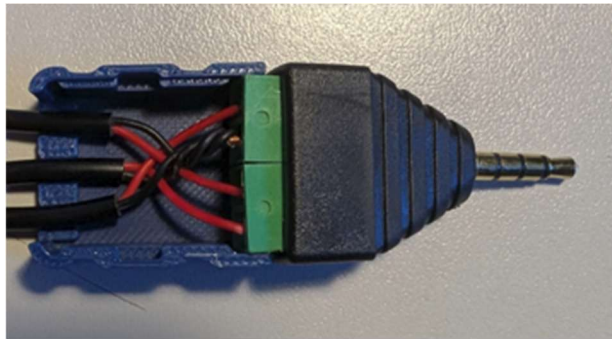
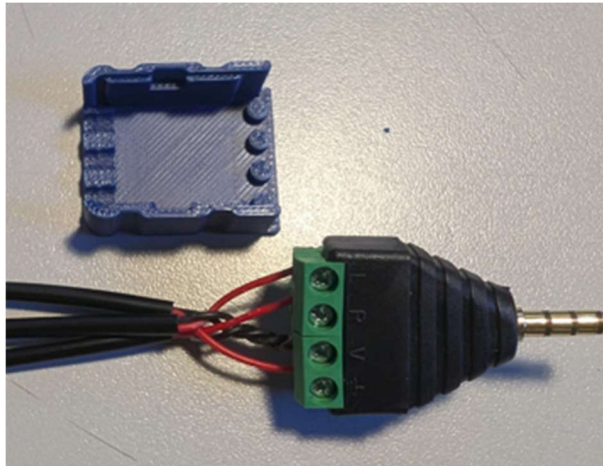


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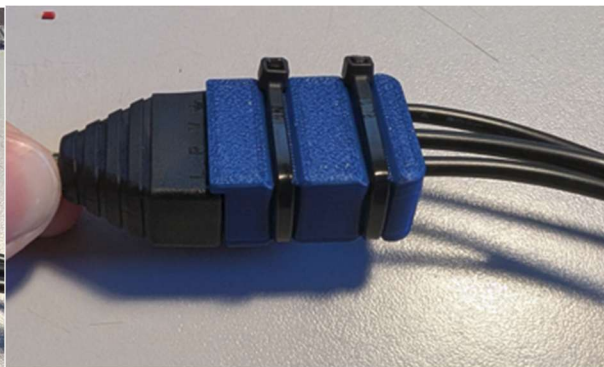
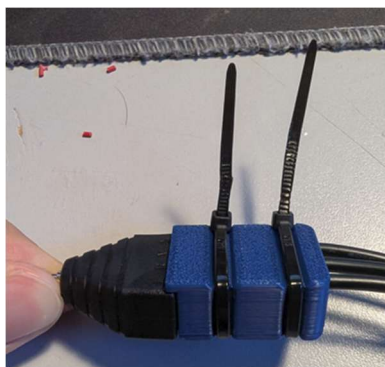
Step B-04: Assemble the strain relief

Take the strain relief base (B03) and place the four pegs on one side into the four screw holes in the terminal jack. Line up the three cables with the three slots on the base, then place the strain relief top (B04) on top.



Step B-05: Cable tie the strain relief together

Take two cable ties (B05) and wrap them around the strain relief enclosure, lying them into the slots in the enclosure. Tighten them as tight as possible, then trim them flush.



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Part C: Flashing Firmware to Shrub Hub

Part C: Required Tools and Supplies

- Computer that is able to install Arduino IDE and other Arduino libraries to flash firmware

Step C-01: Connect Shrub Hub to the Computer

Plug the Shrub hub into a USB port on the computer. If the computer does not have a USB A port, an adapter may be required to connect the Hub to the computer.

Step C-02: Setup Arduino IDE on Computer

1. Download Arduino IDE for your operating system at <https://www.arduino.cc/en/software>.
2. Install the Arduino IDE.

Step C-03: Setup Arduino IDE for TRRS Trinkey

1. Open Arduino IDE.
2. Click on **File -> Preferences**
3. Locate the text field that reads **Additional Boards Manager URLs** beside it.
4. Copy and paste the following link into the field as a new line:

https://adafruit.github.io/arduino-board-index/package_adafruit_index.json

5. Click on **OK**.
6. Restart the Arduino IDE by closing and opening the program.
7. Open the **Boards Manager** option from the **Tools -> Board -> Boards Manager...** menu.
8. Search for "Arduino SAMD" and select "Arduino SAMD Boards" by Arduino. Make sure you select the exact board manager name stated as there may be some with similar names that will not work.
9. Click **Install** to install the board.
10. Still in the **Boards Manager** menu, search for "Adafruit SAMD" and select "Adafruit SAMD Boards" by Adafruit. Make sure you select the exact board manager name stated as there may be some with similar names that will not work.
11. Click **Install** to install the board.

Step C-04: Install Libraries

1. Go to the **Tools -> Manage Libraries...** menu.
2. For each library in the table below, search for the name. Ensure the Author and Version is correct, then install the library. If prompted to install any dependent libraries, click **OK**.

Library Name	Author	Version
Keyboard.h	Arduino	1.0.2
Adafruit NeoPixel.h	Adafruit	1.12.5
Adafruit_SleepyDog.h	Adafruit	1.6.5

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Step C-05: Setup Local Code Directory

1. Download the Firmware_Files from the GitHub Repository

https://github.com/makersmakingchange/Shrub-Hub/blob/main/Build_Files/Firmware_Files/TRRS_Switch_Control.zip

2. Extract / unzip the folder to a known location on your computer.
3. Confirm you have the following folder structure:
 - TRRS_Switch_Control
 - o TRRS_Switch_Control.ino
 - o my_inputs.h
 - o my_stuff.h

Step C-06: Flash Firmware to the Shrub Hub

1. Open the TRRS_Switch_Control.ino with Arduino IDE.
2. Select Adafruit TRRS Trinkey M0 from the **Tools -> Board -> Adafruit SAMD** menu
3. Plug the Shrub Hub into the computer.
4. Select the correct port from the **Tools -> Port** menu. It will say COMX (Adafruit Trinkey M0 (SAMD21)) where COMX is replaced with the number of the port that it is connected to, i.e. COM7
5. Verify and upload to code by clicking the arrow in the top-left corner of the Arduino program.

Testing

Once the Shrub Hub is programmed, connect switches to each of the 3.5mm ports. Test that each switch produces the correct output, and that holding any switch for more than 4 seconds switches modes.

Mark each switch port with the correct switch input number.

Input	Output			
	Mode 1 (Teal)	Mode 2 (Purple)	Mode 3 (Yellow)	Mode 4 (Green)
Switch 1	Enter	F1	1	F13
Switch 2	Space	F2	2	F14
Switch 3	Tab	F3	3	F15
Switch 1 + 2	Backspace	F4	4	F16
Switch 2 + 3	Escape	F5	5	F17
Switch 1 + 3	Currently Unused	Currently Unused	Currently Unused	Currently Unused
Switch 1 + 2 + 3	Currently Unused	Currently Unused	Currently Unused	Currently Unused

Troubleshooting

- A switch input is constantly being held/the Shrub Hub constantly cycles through modes
 - o Check the cables in the splitter for a short between two of the cables

